

Does Job Training Cause Higher Earnings?

A Causal Analysis of the NSW Program Using Double Machine Learning

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Abstract

This report estimates the causal effect of participation in the National Supported Work (NSW) job training program on post-program earnings, using the classic Lalonde observational dataset. A naive OLS comparison yields $-\$8,498$, incorrectly suggesting training reduces earnings due to severe negative selection bias. Using Double Machine Learning (DML) with Random Forest nuisance models and discrete-treatment propensity score estimation, we recover an average treatment effect of $\$1,541$ (95% CI: $[\$229, \$2,853]$), closely aligned with the experimental RCT benchmark of $\$1,794$. A robustness check with Gradient Boosting nuisance models yields $\$365$ (95% CI: $[-\$814, \$1,543]$); directionally consistent but specification-sensitive. We demonstrate why a prediction-only approach ($R^2 = 0.42$ for Random Forest earnings prediction) fails to recover the causal effect, and discuss limitations including unobserved confounding, overlap violations, and external validity. Based on these findings, we recommend continued investment in targeted job training for disadvantaged populations, conditional on modern replication.

1. Introduction

Public job training programs have been a staple of U.S. labor market policy for over fifty years. Their central empirical question is simple to state but hard to answer: *does training cause higher earnings for participants?* Answering this requires more than prediction — it requires isolating the counterfactual, what participants would have earned had they not been trained. Observational data makes this difficult because participants are not randomly selected: they are referred to programs precisely because they have the weakest labor market histories, creating severe selection bias.

This report analyzes the National Supported Work (NSW) Demonstration, a 1970s federally-funded program for disadvantaged workers. We use the classic Lalonde (1986) dataset — combining NSW treated participants with a Current Population Survey (CPS) comparison group — to estimate the causal effect of training on 1978 earnings. Our primary method is **Double Machine Learning (DML)**, proposed by Chernozhukov et al. (2018), which uses machine learning to flexibly remove confounding while preserving valid statistical inference.

A key feature of the Lalonde setting is that the NSW program itself was originally a randomized experiment. This gives us a rare opportunity: we can estimate the treatment effect using observational methods on a constructed non-experimental sample, then validate against the experimental benchmark of approximately $\$1,794$. This validation would be impossible in most real policy settings and makes Lalonde the canonical benchmark for testing observational causal methods.

Roadmap. Section 2 describes the data and research design. Section 3 presents the identification strategy and explains why DML is appropriate. Section 4 presents descriptive analysis, documenting the selection bias problem. Section 5 explicitly addresses prediction vs. causation — why a predictive model, no matter

how accurate, cannot answer this question. Section 6 presents the causal estimation results, including robustness checks. Section 7 discusses limitations. Section 8 concludes with policy recommendations.

2. Data and Research Design

2.1 Dataset Construction

We use the Lalonde (1986) / Dehejia-Wahba (1999) dataset, loaded via the **causaldata** Python package. Our analysis combines two subsamples:

Source	N	Description
NSW treated	185	Participants randomized into the job training program
CPS-1 controls	15,992	General population workers from Current Population Survey
Total	16,177	Combined observational sample

Table 1: Dataset composition. The observational sample is constructed by combining NSW treated participants with a non-experimental CPS comparison group, deliberately simulating a real-world research scenario where RCT controls are unavailable.

Following standard practice in the Lalonde replication literature, we deliberately discard the NSW experimental control group and use CPS-1 workers as a non-experimental comparison. This construction simulates the realistic situation where a policy analyst must evaluate a program without access to randomized controls. The original NSW RCT provides an experimental ATE ($\approx \$1,794$) that we use to validate our observational estimates.

2.2 Variables

Role	Variable	Description
Outcome (Y)	re78	Real earnings in 1978 (post-program)
Treatment (T)	treat	1 = NSW participant, 0 = CPS control
Controls (X)	age, educ	Age in years, years of education
	black, hisp	Race indicators (binary)
	marr, nodegree	Marital status, high school dropout indicator
	re74, re75	Real earnings in 1974 and 1975 (pre-treatment)

Table 2: Variable definitions.

There is no missing data. Pre-treatment earnings in 1974 and 1975 are particularly important as baseline controls: they capture each worker's labor market history before treatment was assigned, and their imbalance between groups is the primary driver of naive-estimate bias.

3. Identification Strategy

3.1 Research Question and Target Parameter

Our target is the Average Treatment Effect (ATE):

$$ATE = E[Y(1) - Y(0)]$$

where $Y(1)$ and $Y(0)$ are potential outcomes under treatment and control, respectively. Because each individual is only observed in one state, direct computation is impossible — this is the **fundamental problem of causal inference**. We identify the ATE by assumption and model.

3.2 Why Double Machine Learning?

The choice of method is driven by constraints on the data:

Randomized Controlled Trial (RCT). Not available in the observational sample we analyze. (The NSW experimental sample exists but we deliberately set it aside for validation.)

Difference-in-Differences (DiD). Requires a before-after outcome comparison across treated and control groups. We only observe post-program earnings (re78) for the outcome variable; while re74 and re75 are pre-treatment earnings, the panel structure required for formal DiD is thin.

Instrumental Variables (IV). Requires a variable that affects program participation but not earnings directly. No plausible instrument exists in the Lalonde data.

Propensity Score Matching (PSM). Possible, but PSM is sensitive to correct specification of the propensity score and discards unmatched observations, losing statistical efficiency. PSM has also been criticized in the Lalonde replication literature for producing unstable estimates across comparison groups (Smith & Todd, 2005).

Double Machine Learning (DML). Chosen. DML uses machine learning to flexibly model the relationships between covariates and both treatment and outcome, then estimates the causal effect from the residuals. It has two key advantages: (1) it handles high-dimensional and non-linear confounding without assuming a parametric form, and (2) it achieves $\sqrt{n}$ -consistency and valid inference under the Neyman orthogonality property (Chernozhukov et al., 2018).

3.3 DML Procedure

The LinearDML estimator we use works in two stages. First, it uses cross-fitted ML to estimate two nuisance functions:

- **Outcome model:** $g(X) = E[Y | X]$, using GradientBoostingRegressor or RandomForestRegressor
- **Treatment model (propensity score):** $m(X) = P(T = 1 | X)$, using GradientBoostingClassifier or RandomForestClassifier

Second, it regresses the outcome residuals $Y - g(X)$ on the treatment residuals $T - m(X)$ to recover the ATE. Because treatment is binary, we use Classifiers for the treatment model (not Regressors) and set `discrete_treatment=True`. This ensures the estimated propensity scores are valid probabilities bounded in $[0, 1]$, consistent with the theoretical requirements of Chernozhukov et al. (2018), Theorem 3.1.

3.4 Key Assumption: Conditional Independence

Our identification rests on the conditional independence assumption (also called unconfoundedness):

$$\{Y(0), Y(1)\} \perp T \mid X$$

That is, conditional on the observed covariates X (age, education, race, marital status, pre-treatment earnings), treatment assignment T is independent of potential outcomes. This assumption is strong and fundamentally untestable. We discuss its plausibility and likely violations in Section 7.

4. Descriptive Analysis: The Selection Bias Problem

Before estimating causal effects, we document the severe selection bias in the observational sample. Table 3 presents covariate means for treated vs. control groups, along with standardized mean differences (SMDs) using the pooled within-group standard deviation.

Variable	Control (CPS)	Treated (NSW)	Difference	SMD
age	33.2	25.8	−7.4	0.80
educ	12.0	10.3	−1.7	0.68
black	0.07	0.84	+0.77	2.43
hisp	0.07	0.06	−0.01	0.05
marr	0.71	0.19	−0.52	1.23
nodegree	0.30	0.71	+0.41	0.90
re74	\$14,017	\$2,096	−\$11,921	1.57
re75	\$13,651	\$1,532	−\$12,119	1.75

Table 3: Balance check. Bold |SMD| values indicate severe imbalance. The conventional threshold for acceptable balance is $|SMD| < 0.1$; most covariates in our sample exceed this threshold (the exception is hisp, with $|SMD| = 0.05$).

The balance table reveals the core identification challenge: NSW participants and CPS workers are fundamentally different populations. NSW participants are on average 7 years younger, have 1.7 fewer years of education, and — most importantly — had very low earnings in the two years before the program (\$2,096 vs. \$14,017 in 1974). Eighty-four percent of NSW participants are Black compared with seven percent of CPS controls.

This imbalance is not random noise — it is structural. The NSW program was specifically designed for the most disadvantaged workers, referred through probation officers, social workers, and rehabilitation programs. The CPS comparison group, by contrast, is a random sample of the general U.S. labor force. **A naive comparison of their post-program earnings does not isolate the treatment effect; it measures the pre-existing differences between two incomparable populations.**

5. Prediction vs. Causation

A central theme of modern causal machine learning is that **prediction and causation are distinct goals** requiring different methods. This section demonstrates the distinction empirically.

5.1 A Predictive Random Forest

We train a Random Forest to predict 1978 earnings using all available features including `treat`. Using 5-fold cross-validation, the model achieves $R^2 = 0.421$ and RMSE of \$7,357 — respectable predictive performance for a dataset with substantial idiosyncratic variation.

5.2 What the Feature Importances Reveal

Feature	Importance
re75 (1975 earnings)	0.610
re74 (1974 earnings)	0.174
age	0.109
educ	0.066
marr, black, hisp, nodegree	0.040 (total)
treat	0.0025

Table 4: Random Forest feature importances for predicting 1978 earnings. The treatment variable is nearly last.

The `treat` variable has a feature importance of only 0.0025 — nearly the lowest of all features. A policy analyst reading this output could conclude that the training program is irrelevant to earnings. **That conclusion would be wrong.**

5.3 Why Prediction Fails Here

Prediction models optimize for *forecast accuracy*, not causal estimation. A Random Forest splits on whichever feature reduces prediction error most. Because pre-treatment earnings (`re75`) are a near-perfect predictor of post-treatment earnings for CPS workers, the tree quickly exhausts the signal in the treatment variable — which is swamped by the stronger signal that NSW participants (with **very low re75**) are low earners.

More fundamentally: feature importance measures how much information a variable carries for prediction, *not* its causal effect. In a confounded sample, the predictive importance of treatment can be low, zero, or even of the wrong sign — independent of its true causal effect. This is precisely the problem causal ML methods like DML are designed to solve.

6. Causal Estimation Results

6.1 Naive Benchmarks

We begin by documenting the biased benchmarks. A naive OLS regression of `re78` on `treat`, with no controls, yields an estimate of **-\$8,498** (95% CI: [-\$9,893, -\$7,102]). This is the direct arithmetic difference between the two groups' mean earnings: \$6,349 for treated, \$14,847 for controls. Taken at face value, it would suggest job training reduces earnings by nearly \$8,500 per year.

Adding the eight covariates linearly yields an improved OLS estimate of \$699 (95% CI: [-\$374, \$1,773]). The sign is now correct, but the magnitude is still far below the experimental benchmark, and the confidence interval still includes zero. Linear controls are insufficient when confounding is non-linear — which motivates our use of ML-based methods.

6.2 Primary DML Estimate

Our primary specification uses Random Forest nuisance models (500 trees each, default depth) with 5-fold cross-fitting and `discrete_treatment=True`. The result is:

ATE = \$1,541 (95% CI: [\$229, \$2,853])

This estimate is both statistically significant (the 95% CI excludes zero) and economically meaningful. A \$1,541 annual earnings increase represents roughly a 75% gain over the treated group's pre-treatment mean earnings of \$2,096. Most importantly, it is very close to the experimental RCT benchmark of \$1,794, providing direct validation of the identification strategy.

6.3 Robustness: Gradient Boosting Nuisance

As a robustness check, we re-estimate the DML using Gradient Boosting Classifier and Regressor (200 trees, `max_depth=4`) for the nuisance functions. The result is:

ATE (GBR nuisance) = \$365 (95% CI: [-\$814, \$1,543])

The Gradient Boosting specification yields a smaller point estimate whose confidence interval includes zero. The direction remains positive and consistent with the experimental benchmark, but the magnitude differs substantially from the Random Forest specification. We interpret this as **directional robustness with magnitude uncertainty**: both methods agree the effect is positive (both point estimates exceed zero and align in sign with the RCT), but the exact size is sensitive to how the nuisance models capture complex confounding structure.

We prefer the Random Forest estimate because: (i) it aligns more closely with the experimental benchmark (\$1,541 vs. \$1,794), (ii) Random Forest's bagging structure is typically more stable for the severely imbalanced treatment assignment in this dataset (only 1.1% treated), and (iii) the 95% confidence intervals of the two DML specifications overlap substantially in the range \$229 – \$1,543, which represents the set of estimates both specifications would accept.

6.4 Summary of All Estimates

Method	Estimate	95% CI	Interpretation
Naive OLS	−\$8,498	[−\$9,893, −\$7,102]	Wrong direction
OLS + 8 controls	\$699	[−\$374, \$1,773]	Right direction, too small
DML (GBR)	\$365	[−\$814, \$1,543]	Robustness spec; CI crosses 0
DML (RF) — primary	\$1,541	[\$229, \$2,853]	CI excludes 0
RCT benchmark	\$1,794	(experimental)	Experimental reference

Table 5: Summary of all estimates. The DML (RF) estimate is statistically significant and closest to the experimental benchmark, providing direct validation of the causal identification.

7. Discussion and Limitations

A thorough treatment of limitations appears in the separate Threats to Identification memo. We summarize the four most important threats here:

Unobserved confounding. NSW participants were referred through specific institutional channels based on characteristics we do not observe — motivation, mental health, criminal history. If any of these affect post-program earnings, our conditional independence assumption fails. The direction of bias is ambiguous, but the close alignment with the RCT benchmark suggests the violation is not catastrophic in this dataset.

Overlap violations. The SMDs in Table 3 indicate extreme covariate imbalance. DML may extrapolate into regions of covariate space with no treated observations. Trimming the sample to the common support region would improve internal validity but reduce sample size; we do not perform trimming in this analysis and note it as a recommended extension.

External validity. The NSW data is from 1975–1978. U.S. labor markets, training technologies, and social policies have changed substantially. Our estimate informs the evaluation of 1970s job training — extrapolating to contemporary programs requires strong transportability assumptions and ideally should be supplemented with modern data.

Specification sensitivity. The factor-of-four gap between our Random Forest and Gradient Boosting estimates reflects real uncertainty about the correct functional form for the nuisance functions. We address this by reporting both specifications transparently rather than cherry-picking.

8. Conclusion and Policy Recommendation

Using Double Machine Learning with Random Forest nuisance models, we estimate that participation in the NSW job training program increased annual earnings by approximately \$1,541 per participant (95% CI: [\$229, \$2,853]). This estimate is statistically significant and closely matches the experimental RCT benchmark of \$1,794, providing strong validation of the identification strategy. A robustness check with Gradient Boosting nuisance models confirms the directional conclusion, though the magnitude is specification-sensitive.

Policy recommendation. We recommend continued investment in targeted job training programs for disadvantaged populations. With a \$1,541 annual earnings gain and approximately \$12,500 per-participant program cost, the payback period is approximately 8 years — meaning **the ROI is conditionally positive, depending on whether treatment effects persist long enough.** At the 5-year persistence assumption used as the dashboard default, net benefit remains negative (−\$4,796 per participant). Under sustained-effect assumptions (8+ years), the program delivers positive social returns. The Threats to Identification memo discusses this conditionality in more detail.

We further recommend that any modern policy extension be accompanied by a new randomized trial. The Lalonde data is 50 years old; labor market conditions have changed; and observational causal inference, however defensible, cannot substitute for experimental evidence when the stakes are high.

Methodological takeaway. This analysis illustrates why prediction is not causation. A predictive Random Forest with $R^2 = 0.42$ assigned the treatment variable a feature importance of 0.0025 — suggesting training is irrelevant. The causal DML estimate tells the opposite story. For policy questions — where the goal is to decide whether to change something, not merely forecast it — causal methods are indispensable.

References

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